Programming Homework: Bayesian Networks and Naïve Bayes

Problem 1: Implementing Inference in Bayesian Networks

Requirements:

Build a program to perform inference on a Bayesian Network for a medical diagnosis problem, using the `pgmpy` library or implementing it from scratch.

a) Construct a Bayesian Network with the following variables and dependency relationships:

- `Smoking`: {Yes, No}
- `LungCancer`: {Yes, No}
- `Cough`: {Yes, No}
- `XRay`: {Normal, Abnormal}

b) Define the Conditional Probability Distributions (CPDs) based on expert knowledge or simulated data.

c) Perform the following types of inference:

- Causal inference: Calculate P(Cough=Yes — Smoking=Yes)
- Diagnostic inference: Calculate P(Smoking=Yes — XRay=Abnormal)
- Intercausal inference: Calculate P(LungCancer=Yes — Cough=Yes, XRay=Abnormal)

d) Compare the inference results with manual calculations from the joint distribution.

e) Visualize the Bayesian Network and explain the meaning of the inference results.

Hints:

- Use `BayesianNetwork` from `pgmpy.models`
- Use `VariableElimination` or `BeliefPropagation` for inference
- Use `networkx` and `matplotlib` for visualization

Problem 2: Text Classification with Naïve Bayes

Requirements:

Implement a Naïve Bayes model for text classification (spam/ham) and compare it with the existing implementation in scikit-learn.

a) Implement a Naïve Bayes model from scratch with the following features:

- Text preprocessing: tokenization, vocabulary building
- Calculate prior probabilities P(Spam) and P(Ham)
- Calculate likelihoods P(Word—Spam) and P(Word—Ham) with Laplace smoothing
- Predict the class for new text samples

b) Compare with `MultinomialNB` from scikit-learn on the same dataset:

- Accuracy
- Precision, Recall, F1-score
- Training and prediction time

c) Experiment with different text preprocessing techniques:

- TF-IDF instead of bag-of-words

- Stopword removal
- Stemming/Lemmatization

d) Evaluate the effect of vocabulary size on model performance.

e) Analyze the most significant words for spam/ham classification.

Hints:

- Use the `spam.csv` dataset from Kaggle or the built-in dataset from scikit-learn
- Use `CountVectorizer` or `TfidfVectorizer` from scikit-learn
- Use `MultinomialNB` for model comparison
- Evaluate using `classification_report` and `confusion_matrix`

References

- pgmpy Documentation: https://pgmpy.org/
- Scikit-learn Naïve Bayes: https://scikit-learn.org/stable/modules/naive_bayes.html
- Lecture Slides: 1.BayesNet - Intro.pdf and 5.NaiveBayes.pdf